

Cash Reconciliation Exception Detection and Triage Automation

Rule-based reconciliation core with AI-assisted exception support

WORKFLOW Cash reconciliation in a control-sensitive operations setting	CORE LOGIC Deterministic matching, break isolation, and exception classification
AI ROLE Post-detection support for explanation, next-step guidance, and draft notes	OUTPUTS Matched transactions, exceptions queue, AI-enhanced queue, and summary report

Overview

This project is a control-aware automation case built around a real and widely recognized finance operations problem: cash reconciliation remains a manual, exception-heavy, and control-sensitive workflow in many environments.

The solution uses a rule-based reconciliation core to compare external bank statement activity against an internal cash ledger, classify outcomes, and generate structured review outputs. An AI-assisted support layer is introduced only after exception detection to help explain likely issues, suggest next steps, and draft analyst notes.

The project is intended to demonstrate automation judgment, workflow analysis, and business-facing solution design in a capital markets-style operations context.

Business Problem

Cash reconciliation is not simply a record comparison exercise. It is an operational control process that requires teams to determine whether records align across amount, reference, and timing, while also distinguishing routine timing-related items from actionable exceptions.

In practice, the burden often comes from:

- manual first-pass review across multiple records
- inconsistent handling of non-standard items
- effort spent organizing exceptions into reviewable categories

- repetitive documentation and follow-up note preparation
- limited visibility into exception workload and priority

This makes reconciliation a strong automation candidate - but only when the workflow is structured carefully and the control boundary is preserved.

Core Design Decision

This is not a fully AI-driven reconciliation tool.

1. RULE-BASED RECONCILIATION CORE	2. AI-ASSISTED EXCEPTION SUPPORT
The control-sensitive part of the workflow remains deterministic and auditable. It is responsible for: • comparing bank statement and internal ledger records • identifying matched items and non-matches • classifying differences into structured exception categories • generating review-ready outputs	AI is applied only after the core reconciliation logic has already identified an exception. It is used to: • explain the likely issue in plain language • suggest a practical next step for analyst review • generate a draft analyst note

This reflects the central design judgment of the case: **in reconciliation, the most important automation decision is knowing where AI should stop.**

Project Outputs

Matched Transactions	Exceptions Queue
A structured output of aligned records, including standard matches and timing-related items identified during reconciliation.	A classified queue of non-matching items with break type, priority, transaction detail, and recommended review action.
AI-Enhanced Exceptions Queue	Summary Report
An enriched version of the exceptions queue with AI-generated exception explanation, suggested next step, and draft analyst note.	A management-facing summary of reconciliation results, including category counts, total amount by category, and visibility into high-priority items.

Together, these outputs show a realistic automation pattern: remove routine work first, structure exception review second, and improve downstream handling without weakening core control logic.

Why This Case Matters

This project is meant to demonstrate more than technical implementation. It shows how a manual operations workflow can be analyzed, structured, and improved through disciplined automation design.

It highlights the ability to:

- understand a control-sensitive operational workflow
- identify realistic scope for automation
- separate deterministic decision logic from AI-supported handling
- convert exception-heavy work into structured review outputs
- communicate a technical solution in business-facing terms

The case uses cash reconciliation as the business process, but its broader purpose is to demonstrate how automation can improve operational efficiency while preserving control clarity and trust.

AI in the Project

AI is intentionally narrow in scope.

It is not used to determine whether transactions truly reconcile, and it does not replace the rule-based control logic of the reconciliation engine. Instead, it is applied downstream from detection to reduce analyst friction through explanation, triage guidance, and documentation support.

This is a feature of the design, not a limitation. In reconciliation, AI is most useful not when it replaces the control decision, but when it improves the handling work that follows it.

PROJECT LINKS

- Interactive Case Presentation: companion file "cash_reconciliation_case_presentation(1).html"
- GitHub Repository: <https://github.com/xiangtinghe616-blip/cash-reconciliation-automation>